Report

Of

Cisco VIP Internship Program

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By:

Roll Number of Student: 25SCS1003000992

Name of Student: Aditya Raj

Batch: 2026-27

1. CANDIDATE'S DECLARATION

I, Aditya Raj, do hereby declare that the internship report titled **"Cisco VIP Internship Program"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Aditya.

Signature
Student Name: Aditya Raj
Roll No.: 25SCS1003000992

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature

Student Name: Aditya Raj

Roll No.: 25SCS1003000992

Table

3. INTERNSHIP COMPLETION CERTIFICATE

Cisco
Networking
Academy

IILM UNIVERSITY

This certificate is awarded to

Aditya Raj

for successfully completing

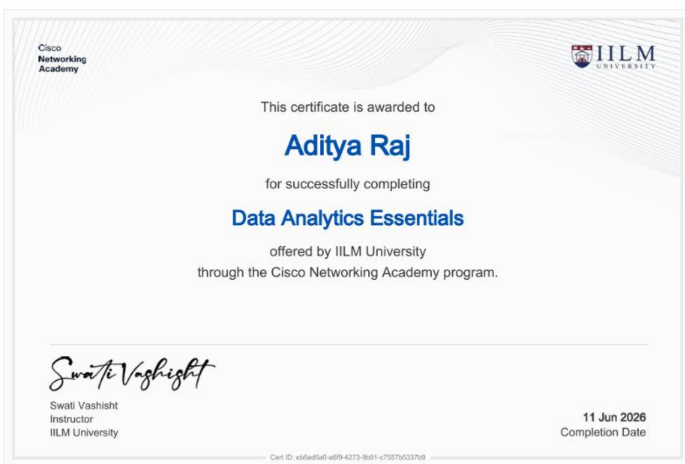
Apply AI: Analyze Customer Reviews

offered by IILM University
through the Cisco Networking Academy program.

Swati Vashisht
Swati Vashisht
Instructor
IILM University

08 Jun 2026
Completion Date

Cert ID: f31ed740-d4c2-4807-8398-fed24168bc84



Cisco
Networking
Academy

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Aditya Raj

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Data Analytics Essentials

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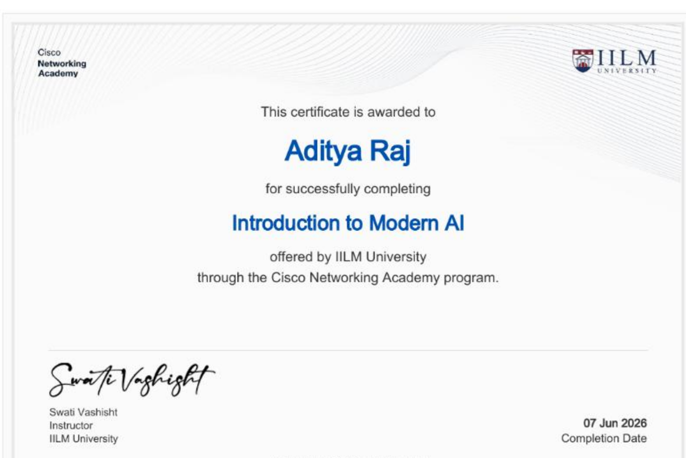
Swati Vashisht
Swati Vashisht
Instructor
IILM University

11 Jun 2026
Completion Date

Cert ID: eb6ad5a0-a6f9-4273-9b91-c7557b5337b9



Cisco
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Introduction to Modern AI

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through the Cisco Networking Academy program.

Swati Vashisht
Swati Vashisht
Instructor
IILM University

07 Jun 2026
Completion Date

Cert ID: 9dcf4644-6b78-4d23-8eb1-b424e2bb04c2

4. Project Description

4.1 Introduction

The Cisco VIP Internship Program provided an opportunity to learn and apply concepts related to Data Analytics, Artificial Intelligence, and networking. The internship was divided into two phases.

In the first phase, I completed three courses: Data Analytics Essentials, Introduction to Modern AI, and Apply AI: Analyze Customer Reviews. These courses helped me understand data preparation, analysis, AI concepts, prompt-based applications, and the use of AI for analyzing real-world data.

In the second phase, I worked on NetSage AI — Applied AI + Network Troubleshooting, an AI-assisted troubleshooting project for Cisco-style networking labs. The project combines AI-based diagnosis with rule-based Python checks and human review to identify possible network faults and suggest appropriate troubleshooting steps.

Overall, the internship helped me connect theoretical concepts with practical applications and improved my understanding of how AI and data-driven techniques can be used to solve technical problems.

4.2 Organization Profile

The **Cisco Networking Academy** is the learning platform through which the courses included in the Cisco VIP Internship Program were provided. During the internship, I completed courses offered by **IILM University through the Cisco Networking Academy program**.
The program provided practical learning in areas such as **Data Analytics, Artificial Intelligence, and AI-based analysis of customer reviews**. These courses formed the first

phase of the internship and provided the knowledge required for the practical project undertaken in the second phase.

IILM University, Greater Noida, through its School of Computer Science and Engineering, is the academic institution under which this internship report is being submitted as part of the B.Tech CSE degree requirements.

This internship structure helped connect academic learning with practical application through a project focused on **AI-assisted network troubleshooting**.

4.3 Problem Statement

The project problem statement provided for your internship identifies a common difficulty faced by junior network engineers: they may know individual networking commands but struggle to connect a reported symptom with the actual root cause. For example, when a PC receives an IP address but cannot reach a server, the fault could be related to VLAN configuration, routing, DHCP, DNS, ACL, or NAT.

The project therefore required the development of an **AI-assisted troubleshooting helper for Cisco-style lab networks**. The system takes network symptoms, Packet Tracer notes, and show command outputs as evidence and suggests the likely fault, relevant OSI layer, next diagnostic command, and an evidence-based fix. A **human reviewer must approve or correct every diagnosis** before it is accepted.

The project was named **NetSage AI** and focused on combining AI-assisted diagnosis with human oversight for safer and more reliable network troubleshooting.

4.4 Project Objectives

The main objective of **NetSage AI** was to develop an AI-assisted troubleshooting system that can help identify common networking problems using available symptoms and network evidence.

The project objectives were to:

1. Develop a dataset of **at least 30 network troubleshooting cases** covering VLAN, gateway, DHCP, DNS, routing, ACL, NAT, and wireless issues.

2. Create structured AI prompts that provide the **root cause, confidence, evidence, next command, and fix steps**.
3. Develop a **Python-based rule checker** for detecting common configuration mistakes independently of AI.
4. Compare AI diagnoses with known solutions and introduce **human review** for every case.
5. Create a dashboard to summarize **issue types, severity, and AI-human agreement**.
6. Maintain a responsible-AI record of cases where the human reviewer corrected or edited the AI response.

The overall aim was to make network troubleshooting more **structured, evidence-based, and human-supervised**.

4.5 Scope of the Project

The scope of **NetSage AI** is to provide an AI-assisted troubleshooting approach for **Cisco-style Packet Tracer lab networks**.

The project covers the following major areas:

1. Network Fault Category:

The system includes troubleshooting cases related to:

- VLAN
- Gateway
- DHCP
- DNS
- Routing
- ACL
- NAT
- Wireless

2. Evidence Based Diagnosis

The AI uses the provided **symptom, topology note, and show-command output** to identify a likely root cause and provide supporting evidence.

3. Diagnostic Assistance

For each case, the system provides:

- Root cause
- OSI layer
- Confidence level
- Evidence
- Next diagnostic command
- Recommended fix steps

4. Rule-Based Validation

A Python-based checker independently looks for common and mechanical configuration mistakes, providing a second signal alongside the AI diagnosis.

5. Human Review

Every AI diagnosis is reviewed by a human before it is considered trustworthy. The reviewer can accept, edit, or reject the diagnosis.

6. Analysis and Reporting

The project includes a dashboard to summarize issue categories, severity, and AI-human agreement.

The project is intended as a **troubleshooting support system for lab-based networking scenarios**, rather than an autonomous system that directly changes network configurations.

4.6 Technologies and Tools Used

The **NetSage AI** project used a combination of Artificial Intelligence, Python-based validation, structured data, and spreadsheet-based analysis.

Technologies and Tools

1. Artificial Intelligence (AI)

AI was used to analyze the networking cases and generate structured troubleshooting diagnoses based on the available evidence.

2. Prompt Engineering

A structured diagnosis prompt was created to make the AI return a consistent JSON response containing the root cause, OSI layer, confidence, evidence, next command, and fix steps.

3. Python

Python was used to develop the **rule checker**, which independently detects common networking configuration errors without using AI.

4. Cisco Packet Tracer / Networking Lab Concepts

The project is designed around Cisco-style Packet Tracer lab troubleshooting and uses networking evidence such as show command outputs.

5. Microsoft Excel / Spreadsheet Tools

Spreadsheet files were used to organize the AI analysis and create the project dashboard containing category, severity, confidence, and AI-human agreement information.

6. CSV Files

CSV files were used to store the troubleshooting cases and the human review records in a structured format.

7. JSON

JSON was used as the standardized format for AI diagnosis responses, making the outputs easier to compare, analyze, and review.

These tools together supported the complete workflow from **case preparation and AI diagnosis to rule-based checking, human review, and final analysis**.

4.7 System Architecture

The **NetSage AI** system follows a sequential workflow in which network information is first collected and then analyzed through multiple stages.

System Workflow

1. Input / Case Data

The system receives a troubleshooting case containing the **symptom, topology note, and show-command output**.

2. AI Diagnosis

The information is provided to the AI using the structured diagnosis prompt. The AI generates a JSON response containing the **root cause, OSI layer, confidence, evidence, next command, and fix steps**.

3. Rule-Based Checker

The same case is checked using the Python rule checker. It independently looks for common configuration problems such as duplicate IP addresses, gateway mismatches, down interfaces, VLAN issues, and missing route/helper configurations.

4. Comparison and Analysis

The AI diagnosis and rule-checker findings are recorded and compared with the expected fault of the case.

5. Human Review

A human reviewer examines the AI response and marks the diagnosis as **Accepted, Edited, or Rejected**. Any correction is recorded in the human review log.

6. Dashboard and Results

The reviewed results are summarized in the project dashboard to show the distribution of issue types, severity, confidence, and AI-human agreement.

Overall Architecture

Case Data → AI Diagnosis → Rule Checker → Comparison → Human Review → Dashboard / Final Result

This architecture ensures that the AI acts as a **decision-support tool**, while the final diagnosis remains under human supervision.

4.8 Methodology

The **NetSage AI** project was developed using a structured troubleshooting and review process.

1. Collect Troubleshooting Cases

A dataset of **30 networking cases** was prepared covering VLAN, Gateway, DHCP, DNS, Routing, ACL, NAT, and Wireless issues. Each case included the symptom, topology note, show-command output, expected fault, OSI layer, concept, and severity.

2. Create the AI Prompt

A structured prompt was designed to ensure that the AI returned a consistent JSON response containing the **root cause, OSI layer, confidence, evidence, next command, and fix steps**.

3. Run AI Diagnosis

Each case was provided to the AI for diagnosis. The AI response was then stored for further comparison and review.

4. Perform Rule-Based Checking

A Python script independently checked the show-command output for common configuration mistakes such as duplicate IP addresses, invalid subnet masks, gateway mismatches, down interfaces, VLAN issues, and missing route/helper configurations.

5. Compare the Results

The AI diagnosis and rule-checker findings were compared with the known expected fault for each case.

6. Human Review

Every AI response was reviewed by a human and marked as **Accepted, Edited, or Rejected**. Corrections were documented where necessary, following the human-review requirement of the project.

7. Prepare Final Analysis

The reviewed results were organized into the dashboard to present the project findings, including issue categories, severity, and AI-human agreement.

Overall Methodology:

Case Collection → Prompt Design → AI Diagnosis → Rule Checking → Result Comparison → Human Review → Dashboard & Final Analysis

4.9 Expected Outcomes

The expected outcomes of the **NetSage AI** project are:

1. Structured Troubleshooting Support

The system provides a structured diagnosis for networking problems instead of relying only on individual commands or manual guesswork.

2. Evidence-Based AI Responses

The AI diagnosis is based on the provided symptoms, topology information, and show-command evidence.

3. Independent Rule Checking

The Python rule checker provides a separate deterministic check for common configuration mistakes, helping to support the AI diagnosis.

4. Human-Supervised Decisions4. Human-Supervised Decisions

Every AI response is reviewed by a human before it is accepted, ensuring that the system does not make the final decision independently.

5. Analysis of AI Performance

The project records AI-human agreement and documents cases where the AI response needed correction.

6. Practical Learning

The project provides practical experience in **Artificial Intelligence, prompt design, Python, networking troubleshooting, data organization, and responsible AI use**.

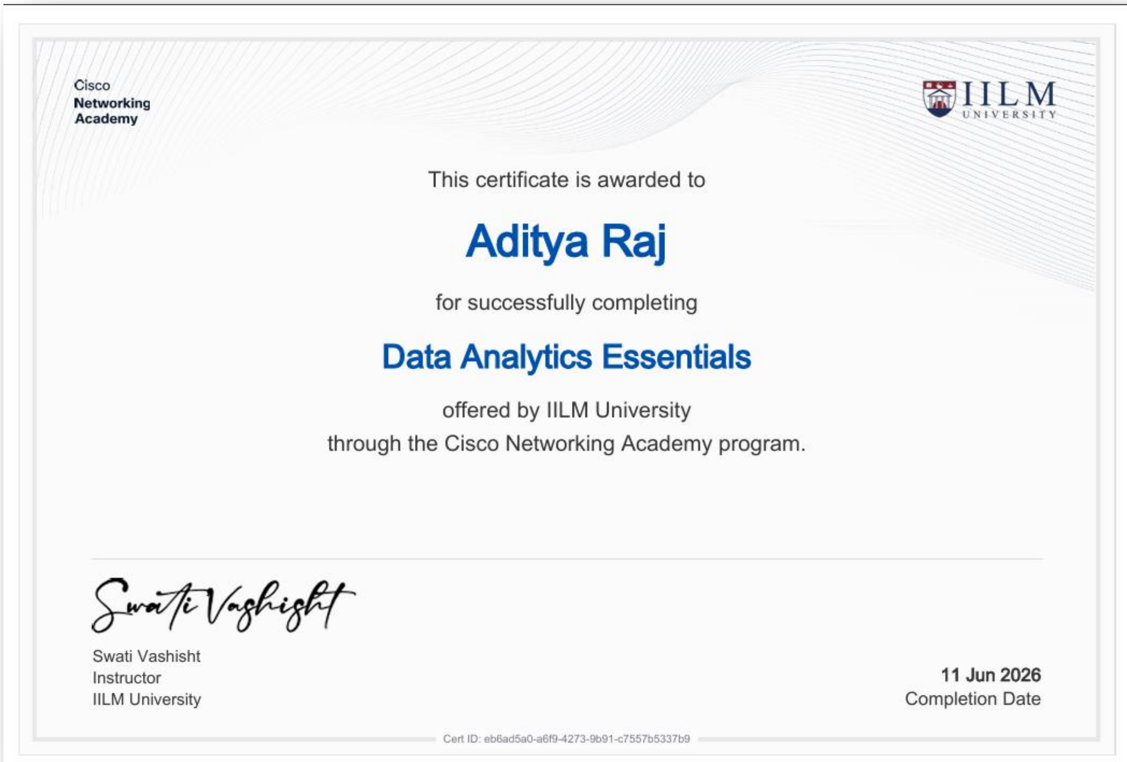
Project Results

The completed project contains **30 troubleshooting cases across 8 fault categories**. The human review recorded **25 accepted responses and 5 edited responses**, with no rejected responses, resulting in an **83.3% AI-human agreement rate**.

The rule checker independently detected at least one deterministic issue in **7 of the 30 cases** during its actual run.

4.10 Certificates of Completion and Communication Proof

1. Data Analytics Essentials



2. Introduction to Modern AI

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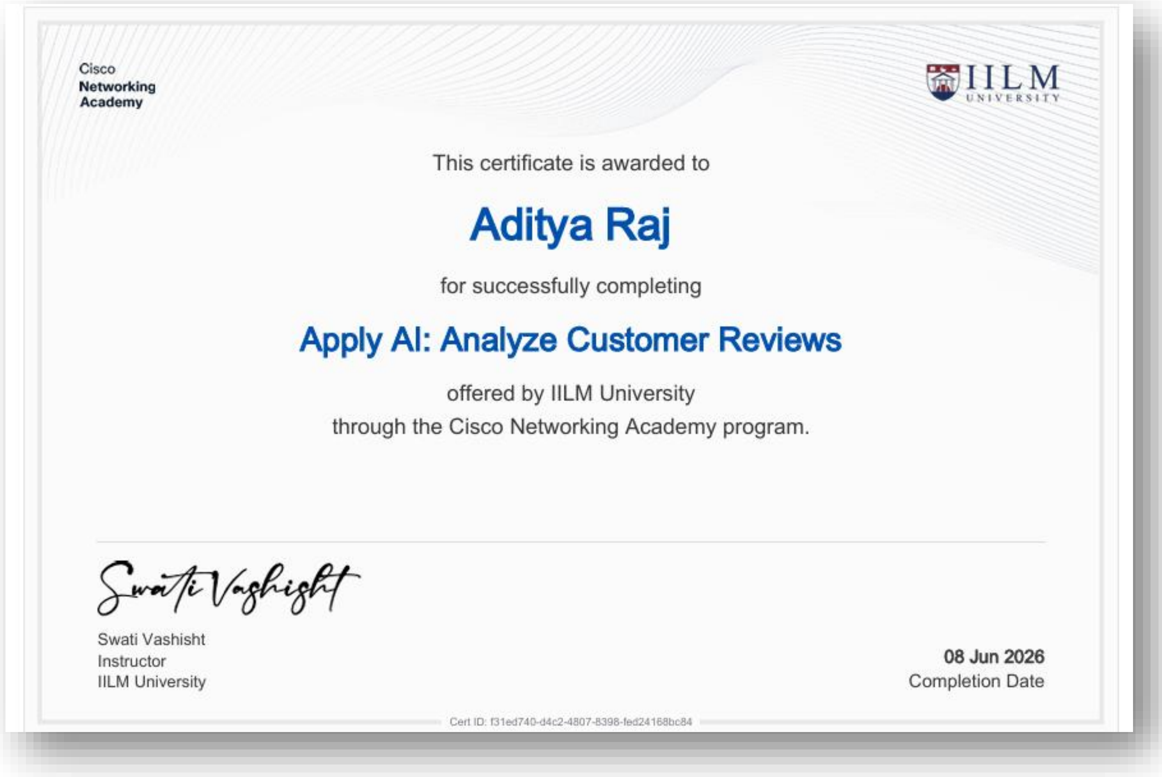
Cert ID: 9dcf4644-6b78-4d23-8eb1-b424e2bb04c2

3. Apply AI: Analyze Customer Reviews



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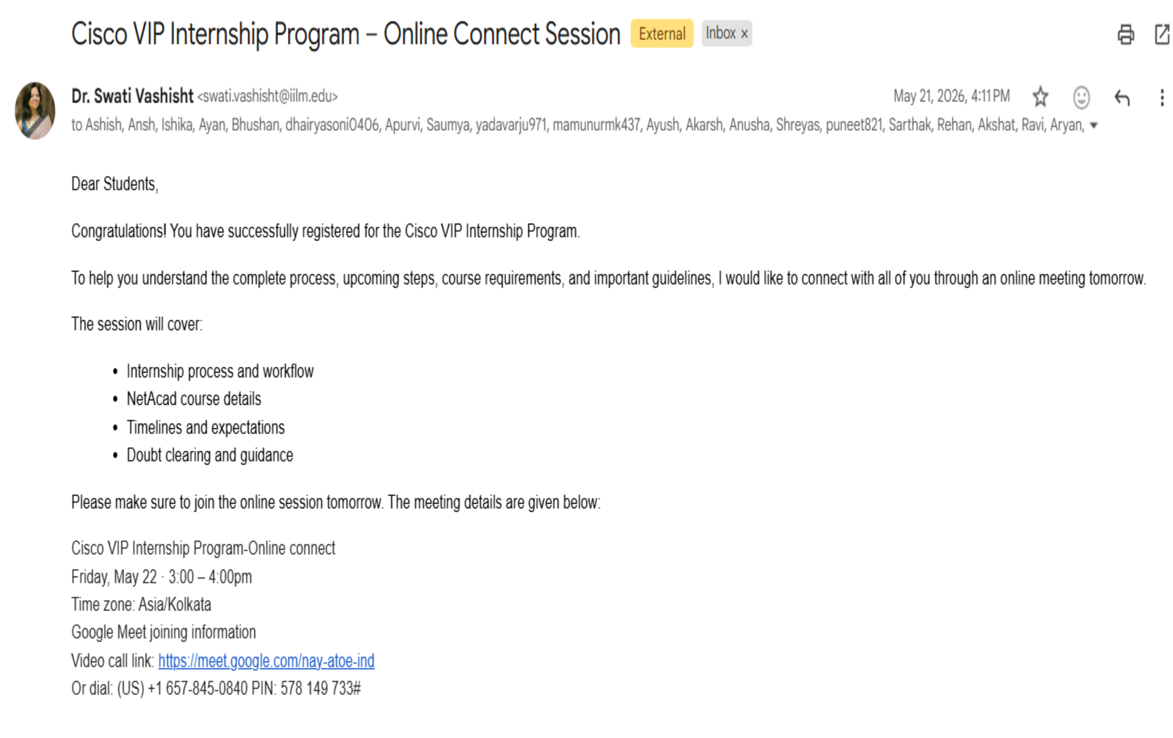
08 Jun 2026
Completion Date

Cert ID: f31ed740-d4c2-4807-8398-fed24168bc84

4. E-mail Proof



Cisco VIP Internship Program – Online Connect Session External Inbox ×

Dr. Swati Vashisht <swati.vashisht@iilm.edu> May 21, 2026, 4:11 PM
to Ashish, Ansh, Ishika, Ayan, Bhushan, dhairyasoni0406, Apurvi, Saumya, yadavarju971, mamunurmk437, Ayush, Akarsh, Anusha, Shreyas, puneet821, Sarthak, Rehan, Akshat, Ravi, Aryan,

Dear Students,

Congratulations! You have successfully registered for the Cisco VIP Internship Program.

To help you understand the complete process, upcoming steps, course requirements, and important guidelines, I would like to connect with all of you through an online meeting tomorrow.

The session will cover:

- Internship process and workflow
- NetAcad course details
- Timelines and expectations
- Doubt clearing and guidance

Please make sure to join the online session tomorrow. The meeting details are given below:

Cisco VIP Internship Program-Online connect
Friday, May 22 · 3:00 – 4:00pm
Time zone: Asia/Kolkata
Google Meet joining information
Video call link: https://meet.google.com/nay-atoe-ind
Or dial: (US) +1 657-845-0840 PIN: 578 149 733#

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5. Bibliography / References

For your report, keep the references limited to the materials actually used for the internship and project:

1. **Cisco Networking Academy**, *Data Analytics Essentials*, course material, IILM University, 2026.
2. **Cisco Networking Academy**, *Introduction to Modern AI*, course material, IILM University, 2026.
3. **Cisco Networking Academy**, *Apply AI: Analyze Customer Reviews*, course material, IILM University, 2026.
4. **Cisco Networking Academy / IILM University**, *Project 2 | Applied AI + Network Troubleshooting — NetSage AI*, project problem statement, 2026.
5. **NetSage AI Project Files**, including cases.csv, diagnose_prompt.md, rule_checker.py, human review log, AI analysis, and dashboard, 2026.
6. **IILM University, School of Computer Science and Engineering**, *Internship Report Format Updated 2026–27*, report guidelines.